THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 3

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of the following metals does not react with dilute HCl? 1

(a) zinc **(b)** copper **(c)** iron **(d)** magnesium

2. Soap does not lather well in hard water because of: 1

(a) low temperature **(b)** calcium and magnesium salts forming scum

(c) sodium chloride **(d)** acidity

3. In the electrolysis of water, the volumes of hydrogen and oxygen collected are in the ratio: 1

(a) 1 : 1 **(b)** 1 : 2 **(c)** 2 : 1 **(d)** 3 : 1

4. Methyl orange in acidic solution is: 1

(a) blue **(b)** colourless **(c)** red **(d)** yellow

5. In the electrolytic refining of copper, the electrolyte is: 1

(a) NaCl solution **(b)** dilute HCl

(c) molten copper **(d)** acidified copper sulphate solution

6. The number of isomers of butane is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

7. Magnesium ribbon is cleaned before burning in air in order to: 1

(a) convert it into a powder

(b) remove the protective oxide layer from its surface

(c) make it burn with a green flame

(d) make it heavier

8. Which type of medicine is used for treating indigestion? 1

(a) antacid **(b)** antibiotic **(c)** antiseptic **(d)** analgesic

9. Food cans are coated with tin and not with zinc because: 1

(a) zinc is cheaper than tin **(b)** zinc has a higher melting point than tin

(c) zinc is less reactive than tin **(d)** zinc is more reactive than tin

10. The next homologue of C_3H_6 is: 1

(a) C_4H_8 **(b)** C_5H_{10} **(c)** C_4H_{10} **(d)** C_3H_8

11. In $CuO + H_2 \longrightarrow Cu + H_2O$, the substance oxidised is: 1

(a) CuO **(b)** H_2 **(c)** Cu **(d)** H_2O

12. The sting of a bee contains: 1

(a) methanoic acid **(b)** citric acid **(c)** lactic acid **(d)** ethanoic acid

13. Which of the following is an ionic compound? 1

(a) H_2O **(b)** $MgCl_2$ **(c)** CCl_4 **(d)** CH_4

14. Esters are formed when a carboxylic acid reacts with: 1

(a) an alcohol **(b)** an alkane **(c)** a base **(d)** a metal

15. $CaCO_3 \longrightarrow CaO + CO_2$ is an example of: 1

(a) thermal decomposition **(b)** displacement

(c) electrolytic decomposition **(d)** photochemical decomposition

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Write the equation for the formation of sodium chloride showing electron transfer. 2

17. What is esterification? Write the equation for ethanol and ethanoic acid. 2

18. Write the balanced chemical equation for the reaction of hydrogen with chlorine to give hydrogen chloride. 2

19. While diluting an acid, why is it recommended that the acid be added to water and not water to acid? 2

20. Why is sodium kept immersed in kerosene oil? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Write the equations for the reaction of 3

(a) iron with steam

(b) calcium with water.

22. Write the molecular formula and structure of ethene. Why is it unsaturated? 3

23. Differentiate between a combination and a decomposition reaction with one example each. 3

24. How is bleaching powder prepared? State two uses. 3

OR

Why is mercury obtained from cinnabar simply by heating? Write the equations. 3

25. What happens when ethanoic acid reacts with sodium hydroxide and with sodium carbonate? Write equations. 3

OR

In the electrolysis of water, why is the volume of gas collected in one test tube double that in the other? Name the gases. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe what happens when a magnesium ribbon is burnt in air. Write the equation, name the type of reaction and state the nature of the product. 5

OR

Describe the activity to show that acids and glucose solutions behave differently in conducting electricity. State the conclusion. 5

27. Arrange Na, Mg, Al, Zn, Fe, Pb, Cu, Ag, Au in order of reactivity. Explain what happens when each class of metal reacts with water and acids. 5

OR

Explain structural isomerism with isomers of pentane: draw formulae and name them. 5

END OF QUESTION PAPER **Total: 50 Marks**

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